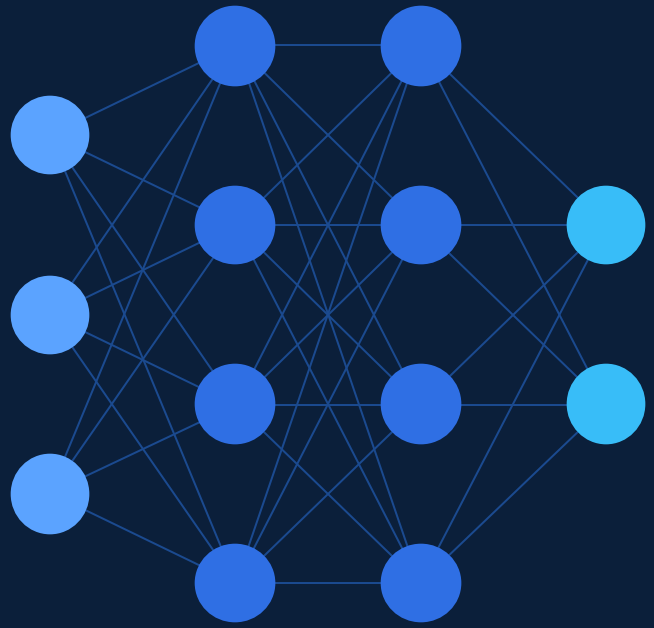
Machine Learning vs Deep Learning

Samarth Rajput

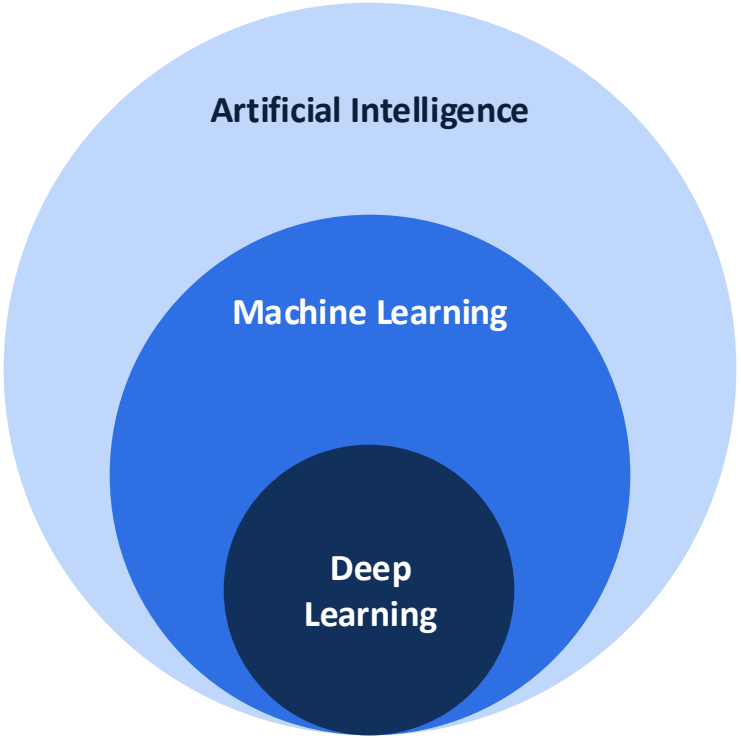
ADIT, Anand

Subject: AI for Everyone



What is Artificial Intelligence?

- AI: machines that do tasks needing human intelligence.
- ML: a part of AI that learns from data.
- DL: a part of ML that uses neural networks.
- Each one sits inside the one before it.



What is Machine Learning?

- ML lets computers learn from data, without fixed rules.
- Example: a spam filter learns which emails are junk.

EXAMPLE: SPAM FILTER



3 TYPES OF ML

1 Supervised

Learns from labeled examples

2 Unsupervised

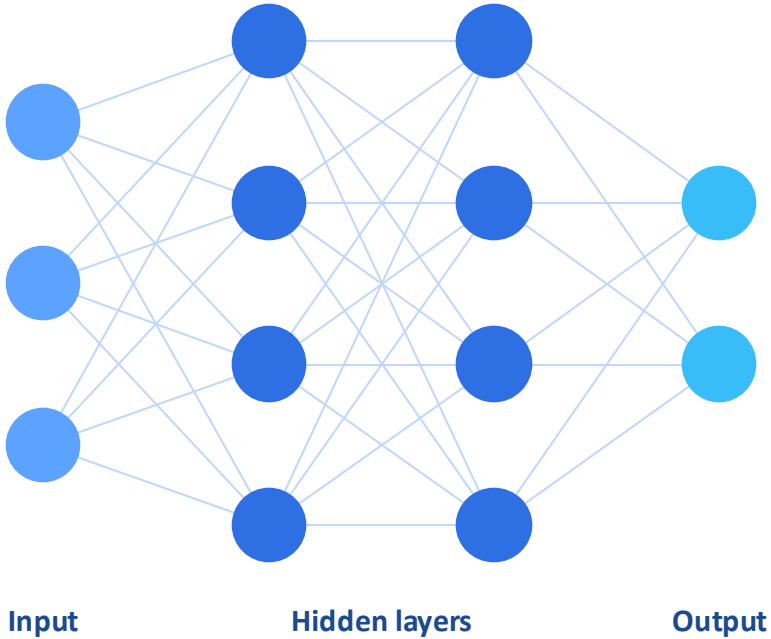
Finds hidden patterns in data

3 Reinforcement

Learns by rewards and penalties

What is Deep Learning?

- DL is ML that uses neural networks with many layers.
- A neural network is connected nodes, like brain cells.
- Each layer learns something more complex than the last.
- Example: face unlock and voice assistants like Siri.



ML vs DL: Comparison

Feature	Machine Learning	Deep Learning
Data needed	Works with small data	Needs very large data
Training time	Minutes to hours	Hours to days or weeks
Hardware needed	Normal CPU is enough	GPU is usually needed
Feature extraction	Manual (done by humans)	Automatic (model learns it)
Accuracy	Good on simple tasks	Higher on complex tasks
Examples	Spam filter, price prediction	Face recognition, voice assistants

Real-World Applications

ML

Machine Learning

Spam filters, price prediction, product recommendations

DL

Deep Learning

Face recognition, voice assistants, self-driving cars

When to use which?

Use ML when: data is small and results must be explainable.

Use DL when: data is huge and the problem is complex.

Conclusion

1

AI contains ML, and ML contains DL.

2

ML works with less data; DL needs more data and GPUs.

3

Pick the method based on your data, task and hardware.

Thank you! Questions?